

Subandhu Agravanshi

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SUMMARY

Generative AI / Applied AI Engineer with **3.5+ years** building production agentic, retrieval and ML systems across enterprise, fintech, gaming and media. Strong across **agentic AI, RAG, retrieval/reranking, multimodal AI, fine-tuning, evaluation, classical ML, data engineering and backend systems**. Takes AI from experimentation to production: serving concurrent users at scale, cost and latency optimization, AI engineering and evals, and measurable business impact.

WORK EXPERIENCE

Pocket FM | *Generative AI Engineer*

Jun 2026 - Present

- Built an **organization-level knowledge search and analysis platform** with specialized agents, sub-agents, skills and tools. Connected enterprise knowledge bases to **Databricks** tables and SQL through Python/OpenWebUI. Teams retrieve grounded evidence, run analysis and generate dashboards, serving concurrent users at scale.
- Improved retrieval with **cross-encoder** query encoding, **Cohere Rerank 3.5**, knowledge-base routing, table/schema metadata and prompt/context optimization. Served Claude Opus-family models through **AWS Bedrock** and **LiteLLM**. Raised retrieval accuracy **30%**. Cut cost from **\$1.30 to \$0.60 per query** (54%).
- Built Text-to-SQL and analytics agents: NL query interpretation, schema/table discovery, metadata-aware context routing, SQL generation and Databricks execution, then analytical output and dashboards. Improved SQL generation precision by **50%**. Reduced manual analysis for organization-level business insight.

Gameopedia | *Generative AI Engineer, Research & Engineering*

Dec 2024 - Jun 2026

- Productionized **video-understanding agentic pipelines** over 3,000+ long-form videos using Gemini 2.5 Flash, CNN/ViT frame analysis, temporal scene segmentation and Python orchestration. Generated **50+ highlights/video** across games, sports, trailers, vlogs and 10+ hour streams, contributing to **+1.2% watch-time**.
- Built Game Marketing Sentiment Intelligence over **20,000+ Steam/Reddit reviews**. Established TF-IDF + Logistic Regression/SVM baselines, scaled to OpenAI Agents across 60+ categories, and deployed through **MCP + Batch API**. Cut inference cost from **\$8 to \$0.30/input** (96%).
- Fine-tuned a **360M Small-LM with LoRA** and evaluated DeepSeek-R1/GPT-3.5 across 60+ runs using loss curves, semantic similarity and qualitative review. Orchestrated taxonomy generation through LangGraph, reaching **78% semantic similarity** and **100+ new insights**.
- Architected an **RLHF-inspired autonomous refinement loop** using Gemini 2.5 Flash + OpenAI Responses API. Automated iterative evaluation and self-correction, improving classification accuracy from **33% to 88%** in three cycles.

Stealth Startup (Fintech) | *Machine Learning Engineer & AI Researcher*

Aug 2024 - Nov 2024

- Audited and structured **1M+ financial prospectuses**. Built **hybrid Graph + LanceDB RAG** with chunking, embedding selection and retrieval benchmarking, then exposed it through an AWS Bedrock/Groq/LangGraph research agent for evidence-backed financial analysis.
- Built financial-document classifiers and compared embedding/ANN retrieval with fine-tuned approaches. Implemented **HYDE, CRAG and Reflexion** POCs plus an **Equity Research Agent** supporting 7+ analyst workflows with Human-in-the-Loop oversight.

Arcesium (D. E. Shaw Group) | *Forward Deployed Software Engineer, Machine Learning Practice*

Mar 2023 - Aug 2024

- Built an end-to-end **tax-character prediction pipeline** using image preprocessing, OCR, feature engineering, CNNs and ensemble classifiers. Integrated it into forward-deployed enterprise processing workflows at **98% accuracy across 250+ document types**.
- Productionized an **LLM test-generation and evaluation framework** using BLEU, ROUGE, semantic similarity and statistical sampling. Compared LLaMA2, Mistral7B and CodeLLaMA, lifting test coverage from **32% to 80%** and generating **24,000 unique test cases**.
- Built a **Perl-to-Python translation evaluation pipeline**. Benchmarked 15+ LLMs on functional accuracy, AST comparison and unit-test pass rates, translating **350 functions** with Vicuna-13B at **75% functional accuracy**.
- Engineered structured/unstructured financial data and LLM pipelines using **AWS Knowledge Bases + LanceDB**. Added **Langfuse observability across 10,000+ flows** and NeMo guardrails for traceability, reliability and generation control.

SKILLS

Agentic AI & LLMs: LLMs, Agentic AI, Multi-Agent Systems, Tool Calling, RAG, Hybrid Retrieval, Rerankers, Cross-Encoders, Vector Search, Knowledge Graphs, MCP, LangGraph, LangChain, OpenAI Agents, OpenWebUI, LiteLLM, Prompt/Context Engineering.

AI Evaluation & Reliability: Retrieval and Agent Evaluation, BLEU/ROUGE/F1/ROC-AUC, Human-in-the-Loop, Guardrails, Langfuse, Structured Outputs, Model Routing, Cost/Latency Optimization.

ML & Deep Learning: PyTorch, TensorFlow, HuggingFace, Scikit-Learn, Transformers, CNNs, ViTs, Temporal Models, LoRA/PEFT, Fine-Tuning, Classification, Regression, Clustering, Recommendations, Ranking, Feature Engineering, Statistical Modeling, A/B Testing.

Engineering & Data: Python, FastAPI, SQL, Databricks, PostgreSQL, Redis, AWS Bedrock, AWS Knowledge Bases, Vertex AI, Docker, Linux, Git, React, Next.js, Vercel.

Multimodal / Media: Runway, Kling, Seedance, Veo, fal.ai, ImageKit.io, ElevenLabs, ASR, Diarization, Image/Video Generation.

SELECTED PROJECTS

Fiction.ai: AI-native filmmaking with Director, Writer, Casting, Art, Cinematography and Editing agents. LangGraph orchestration, **fal.ai**, Runway, Kling, Seedance and Veo, **ImageKit.io** consistency, **ElevenLabs** dialogue, director critique loop.

Research Gap Finder: 9-node **LangGraph** agent over Semantic Scholar, arXiv and OpenAlex. FastAPI + SSE + React, citation-graph analysis, BERTopic. Surfaces **7-15+ gaps** with BibTeX/RIS/DOCX export.

Video Game Recommendation System: Recs over **30,000+ Steam/Stadia reviews** using TF-IDF, collaborative filtering and feature engineering. Evaluated with **precision@k** and coverage.

LifeOS: Production-style AI OS on **Next.js + PostgreSQL/Prisma** (Server Actions, JWT, Vercel). Gateway assistant with propose/apply/undo and audit ledger. Write-through aggregates: **15-25 live joins to 1-3 reads**.

Audio Intelligence Studio: 16-stage FastAPI + Next.js pipeline (VAD, faster-whisper, pyannote, BERTopic, FAISS RAG). GPU-aware cost tracking. Diarization **50% faster**.

EDUCATION

UC Berkeley - Plaksha University | Post Graduate Masters, Applied AI & Product | CGPA: 8.5

2022 - 2023

KIIT | BTech, Computer Science & Engineering | CGPA: 8.01

2018 - 2022

PUBLICATIONS

Springer: "An Analytical Perspective of Machine Learning applied to Cybersecurity" | "Understanding CNN based models applied to interpreting ECG data"